Jigsaw Reading: The Five Drivers of Biodiversity Loss

B2 Level · Environmental Issues for Plants & Animals · 5 Groups · ~60 minutes

Teacher's Notes

This jigsaw reading is built around the **IPBES Five Direct Drivers of Biodiversity Loss**, identified in the 2019 UN Global Assessment Report. Each group becomes the 'expert' on one driver, then regroups into mixed teams to share what they've learnt and fill in the whole picture together.

How to run it

1. Divide the class into 5 expert groups (A–E). Give each student the reading page for their group only. Students read silently (8–10 min), then discuss comprehension questions together in their expert group (10 min). They should agree on answers and check the vocabulary.

2. Re-group into 'mixed' groups of 5 — one expert from A, B, C, D, and E in each team. Each expert teaches their driver (3 min each = 15 min). Everyone fills in the shared grid.

3. Whole-class discussion (10–15 min) using the prompt questions on the grid page.

Printing

Print **enough copies of pages 2–6** so every student in that expert group has their own. Print **page 7 (reconstruction grid) for every student**. Page 8 is the answer key — keep this for yourself.

Language focus (optional)

B2 reporting and hedging language for the re-grouped phase: *According to the text… / It says that… / One key statistic is… / Apparently… / The main point is that…*

Sources (all data verified as of April 2026)

IPBES (2019) Global Assessment Report on Biodiversity and Ecosystem Services · IUCN Red List (2023 update) · Ocean Conservancy / National Geographic studies on marine plastic · IPCC Special Report on the Ocean and Cryosphere · WWF / Polar Bears International · UN / IPBES Invasive Alien Species Report (2023).

GROUP A | Habitat Loss & Land-Use Change

The biggest driver of biodiversity loss on land

Read your text carefully. You will need to explain it later to students from other groups — they won't have read it!

The planet's land and oceans have changed more in the last 50 years than in any other period of human history. According to the IPBES Global Assessment Report (2019), around 75% of the land-based environment and 66% of the marine environment have been significantly altered by human activity. Farming is the biggest single cause: agriculture currently takes up about 50% of all habitable land on Earth, and most of this — around 77% — is used for raising livestock. More than a third of the world's land surface and nearly 75% of freshwater resources are devoted to crops or animals.

When forests are cleared, wetlands are drained, or grasslands are ploughed up, countless species lose the places where they feed, breed, and shelter. Since the 1990s, more than 1.6 million square miles of forest have been cut down — roughly the equivalent of two football pitches every minute. The average abundance of native species in most major land habitats has fallen by at least 20% since 1900, and amphibians such as frogs and salamanders have been hit particularly hard: around 40% are now threatened with extinction.

Urban sprawl and road-building also break habitats into smaller and smaller pieces, so animals cannot move freely to find mates, food, or water. Scientists call this habitat fragmentation, and it makes even protected species far more vulnerable. Without wildlife corridors linking what remains, many populations become too small to survive.

Comprehension — answer on paper or discuss with your expert group:

1. What percentage of the land-based and marine environments have been significantly altered by people?
2. Farming takes up about 50% of habitable land. What is most of that land used for?
3. How quickly have forests been disappearing since the 1990s? (Include the football-pitch comparison.)
4. Which group of animals has been especially badly affected? Give the percentage.
5. What does 'habitat fragmentation' mean, and why is it dangerous for wildlife?

habitat	the natural home of a plant or animal
abundance	how much of something exists; the total amount
urban sprawl	the spread of cities into the countryside
fragmentation	breaking into smaller separate pieces

GROUP B | Overexploitation of Species

Taking more than nature can replace

Read your text carefully. You will need to explain it later to students from other groups — they won't have read it!

When humans take more fish, animals, or plants from the wild than nature can replace, species numbers crash. This is called overexploitation, and it is one of the most direct drivers of extinction. The IUCN Red List now includes more than 42,000 species that are at risk of extinction, and for many of them, overfishing or hunting is the main cause.

The oceans show the clearest example. A 2021 study found that around a third of all sharks and rays — about 391 species — are now threatened with extinction, and overfishing is the only threat for two-thirds of those species. Certain other groups are in even worse shape: 13% of the world's grouper species are threatened, and around 25% of freshwater fish are endangered, partly because of overfishing.

On land, illegal hunting — often for "bushmeat" — has pushed seven primate species further towards extinction in recent years, according to the IUCN. Logging is just as destructive: when ancient forests are cut for timber, everything living in them — from orangutans to tiny orchids — is affected.

The word researchers keep using is **unsustainable**: we are removing species faster than they can recover. Unless quotas, fishing limits, and hunting bans are properly enforced, many iconic animals — tigers, pangolins, bluefin tuna — could disappear within our lifetime.

Comprehension — answer on paper or discuss with your expert group:

1. In your own words, what does 'overexploitation' mean?
2. How many species are currently on the IUCN Red List as being at risk of extinction?
3. What percentage of sharks and rays are threatened with extinction?
4. Give two examples mentioned in the text of illegal hunting or unsustainable fishing.
5. Which word does the text use to describe taking species faster than they can recover?

overexploitation	taking more of something than can be replaced
unsustainable	unable to continue at the current rate
quota	an official limit on how much can be taken
bushmeat	meat from wild animals hunted in forests

GROUP C | Climate Change

A warming planet, a shrinking home

Read your text carefully. You will need to explain it later to students from other groups — they won't have read it!

Rising global temperatures are already rewriting the map of life on Earth. The IPCC warns that if the planet warms by 1.5–2.5°C, around 20–30% of all known species could be lost. Even the 1.1°C of warming we have already caused is doing serious damage.

Coral reefs are among the first ecosystems to collapse. When the sea becomes too warm, corals expel the tiny algae that feed them and turn white — a process called bleaching. If the stress continues, they die. The 2023–2025 global bleaching event has already damaged 83.9% of the world's coral reef area. At just 1.5°C of warming, the IPCC predicts that between 70% and 90% of coral reefs will suffer severe degradation.

At the poles, the picture is just as serious. The Arctic is warming at twice the global rate, and polar bears — which need sea ice to hunt seals — are projected to decline by around 30% by 2050. Other species are not disappearing but migrating: cod and other commercial fish are shifting northwards into cooler waters, which disrupts traditional fishing grounds and local food webs.

Plants and insects are also moving uphill or towards the poles, looking for cooler conditions. For species already living on mountain tops, however, there is nowhere left to go.

Comprehension — answer on paper or discuss with your expert group:

1. How much warming could cause 20–30% of species to be lost, according to the IPCC?
2. What is coral bleaching, and what triggers it?
3. By what year are polar bears expected to decline by 30%, and what is the main reason?
4. What are cod and similar fish doing in response to warmer seas?
5. Why is wildlife on mountain tops especially vulnerable to climate change?

bleaching	losing colour; here, coral turning white and dying
degradation	damage that makes something worse or less valuable
range shift	a change in where a species lives
sea ice	frozen sea water that forms over the Arctic Ocean

GROUP D | Pollution

A silent crisis of plastic and chemicals

Read your text carefully. You will need to explain it later to students from other groups — they won't have read it!

Pollution is one of the fastest-growing threats to wildlife, and plastic waste has become its most visible symbol. Scientists estimate that more than 11 million metric tonnes of plastic enter the ocean every year, and around 8 million new pieces of plastic reach the sea every single day.

The impact on animals is enormous. More than one million seabirds and 100,000 marine mammals are killed by ocean plastic every year. Recent studies have found plastic in 100% of marine turtles, 59% of whales, 40% of seabirds, and 36% of seals that were examined. Turtles are especially at risk because they often mistake floating plastic bags for jellyfish: roughly half of all sea turtles worldwide have swallowed plastic. Seabirds can even starve to death with full stomachs, because the plastic fills up the space where food should go.

The problem is not limited to big items. Microplastics — tiny particles invisible to the naked eye — have now been found in the gut of every one of the world's seven species of sea turtle. These fragments absorb toxic chemicals and pass them up the food chain, eventually reaching the fish that humans eat.

Plastic is only part of the story. Pesticides from farmland, fertilisers that create "dead zones" in coastal seas, and air pollution also threaten wildlife: together these factors put around 57% of freshwater fish species at risk.

Comprehension — answer on paper or discuss with your expert group:

1. How much plastic enters the oceans every year? And every day?
2. What percentage of marine turtles have been found to contain plastic?
3. Why do sea turtles often swallow plastic bags?
4. What are microplastics, and why are they especially dangerous?
5. Besides plastic, what other forms of pollution are mentioned in the text?

microplastics	tiny pieces of plastic, usually smaller than 5 mm
ingest	to eat or swallow something
dead zone	an area of water with no oxygen, where life cannot survive
pesticide	a chemical used to kill insects or weeds

GROUP E | Invasive Non-Native Species

The hidden hitchhikers destroying ecosystems

Read your text carefully. You will need to explain it later to students from other groups — they won't have read it!

An invasive species is a plant or animal that has been introduced — usually accidentally, by ships, planes, or the pet trade — to a place where it does not naturally belong. Without the predators or diseases that keep it in check at home, it multiplies quickly and can destroy native wildlife. According to a 2023 United Nations report, invasive species cost the global economy at least $423 billion every year, and they have played a role in around 60% of all recent plant and animal extinctions.

The examples are striking. After the Second World War, the brown tree snake arrived in Guam, hidden on American cargo ships. In roughly 70 years, it has wiped out nearly every native forest bird on the island, and it regularly causes expensive power outages by climbing onto electric lines. In Australia, the cane toad — introduced in 1935 to control pests in sugar-cane fields — spread uncontrollably. Its skin produces a poison that kills native predators such as quolls and certain snakes when they try to eat it.

Waterways are just as vulnerable. Zebra mussels, originally from Eastern Europe, have invaded North American rivers and lakes, clogging water pipes and filtering so much plankton that native fish starve. US power plants alone spend around $145 million a year trying to control them.

Experts warn that once an invasive species is fully established, complete removal is almost impossible. Prevention — strict border checks, clean ship ballast water, and responsible pet ownership — is the only reliable defence.

Comprehension — answer on paper or discuss with your expert group:

1. What is an invasive species, and how do they usually arrive in a new place?
2. How much do invasive species cost the world each year, according to the UN?
3. What damage has the brown tree snake caused in Guam?
4. Why was the cane toad brought to Australia, and what went wrong?
5. According to the experts quoted, why is prevention more important than removal?

invasive	spreading aggressively into a new area
native	belonging naturally to a place
introduce (a species)	bring a species to a new area, on purpose or by accident
establish	become permanently settled in a new place

GROUP MIXED | Reconstruction Grid — Teach Each Other

You are now in a new group with one expert from each of A, B, C, D, and E. Listen to your teammates and complete the table below. Use your own words — don't just copy the text.

Driver	One key statistic or fact	One affected species	One possible solution
A — Habitat Loss			
B — Overexploitation			
C — Climate Change			
D — Pollution			
E — Invasive Species			

Whole-class discussion

1. Which of the five drivers do you think is the most urgent to tackle? Why?
2. Which driver is most visible in your own country or city right now?
3. Rank the five drivers from 1 (easiest to solve) to 5 (hardest). Compare with a partner — where do you disagree?
4. Which driver affects humans most directly, and in what ways?
5. If you had to choose one species from the texts to save, which one would you pick, and why?

GROUP KEY | Answer Key — Teacher Only

Group A — Habitat Loss & Land-Use Change

1. 75% of the land environment and 66% of the marine environment.
2. Most of it (about 77%) is used for raising livestock.
3. Since the 1990s, more than 1.6 million square miles of forest have been cut down — roughly two football pitches every minute.
4. Amphibians (frogs, salamanders) — around 40% are threatened with extinction.
5. It means breaking habitats into smaller separate pieces, so animals can't move freely to find mates, food or water. It leaves populations too small and isolated to survive.

Group B — Overexploitation of Species

1. Taking more animals, fish or plants from the wild than nature can replace.
2. More than 42,000 species.
3. About a third (around 391 species, or 32.6%).
4. Accept any two: overfishing of sharks and rays; hunting primates for bushmeat; logging ancient forests; unsustainable fishing of tuna/pangolins/tigers.
5. 'Unsustainable'.

Group C — Climate Change

1. Warming of 1.5–2.5°C.
2. When the sea is too warm, corals expel the tiny algae that feed them and turn white. Heat stress is the trigger.
3. By 2050; because they need sea ice to hunt seals, and Arctic sea ice is shrinking twice as fast as the global warming rate.
4. They are moving/migrating northwards into cooler waters.
5. Because as temperatures rise they have to move uphill — but at the top of a mountain there is nowhere left to go.

Group D — Pollution

1. More than 11 million metric tonnes a year; about 8 million new pieces enter the sea every day.
2. 100%.
3. Because floating plastic bags look very similar to jellyfish.
4. Tiny plastic particles invisible to the eye. They absorb toxic chemicals and pass them up the food chain, eventually into human food.

5. Pesticides, fertilisers that create ‘dead zones’, and air pollution.

Group E — Invasive Non-Native Species

1. A plant or animal introduced to a place where it doesn’t naturally belong, usually by accident (ships, planes, the pet trade).
2. At least $423 billion every year.
3. It has wiped out nearly every native forest bird and causes expensive power outages by climbing onto power lines.
4. It was brought in 1935 to control pests in sugar-cane fields, but it spread uncontrollably and its poisonous skin now kills native predators that try to eat it.
5. Once a species is fully established, complete removal is almost impossible, so stopping them arriving in the first place is the only reliable approach.